Asymptotics and Special Functions

Frank W. J. Olver
University of Maryland ,_
College Park, Maryland
and
National Institute of Standards and Technology
Gaithersburg, Maryland

A K Peters
Natick, Massachusetts

CONTENTS

Preface to A K Peters Edition , *xv*

Preface *xvii*

1 Introduction to Asymptotic Analysis

1 Origin of Asymptotic Expansions 1
2 The Symbols ~, *o*, and *O* 4
3 The Symbols ~, *o*, and *O* (continued) 6
4 Integration and Differentiation of Asymptotic and Order Relations 8
5 Asymptotic Solution of Transcendental Equations: Real Variables 11
6 Asymptotic Solution of Transcendental Equations: Complex Variables 14
7 Definition and Fundamental Properties of Asymptotic Expansions 16
8 Operations with Asymptotic Expansions 19
9 Functions Having Prescribed Asymptotic Expansions 22
10 Generalizations of Poincar6's Definition 24
11 Error Analysis; Variational Operator 27
. Historical Notes and Additional References 29

2 Introduction to Special Functions

1 The Gamma Function 31
2 The Psi Function 39
3 Exponential, Logarithmic, Sine, and Cosine Integrals 40
4 Error Functions, Dawson's Integral, and Fresnel Integrals 43
5 Incomplete Gamma Functions 45
6 Orthogonal Polynomials 46
7 The Classical Orthogonal Polynomials 48
8 The Airy Integral 53

9 The Bessel Function $7_v(z)$ 55
10 The Modified Bessel Function $/_v(z)$ 60
11 TheZeta Function 61
Historical Notes and Additional References 64

3 Integrals of a Real Variable

1 Integration by Parts 66
2 Laplace Integrals 67
3 Watson's Lemma 71
4 The Riemann-Lebesgue Lemma 73
5 Fourier Integrals 75
6 Examples; Cases of Failure 76
7 Laplace's Method 80
8 Asymptotic Expansions by Laplace's Method; Gamma Function of Large Argument 85
9 Error Bounds for Watson's Lemma and Laplace's Method §9
10 Examples 92
11 The Method of Stationary Phase 96
12 Preliminary Lemmas 98
13 Asymptotic Nature of the Stationary Phase Approximation 100
14 Asymptotic Expansions by the Method of Stationary Phase 104
Historical Notes and Additional References 104

4 Contour Integrals

1 Laplace Integrals with a Complex Parameter 106
2 Incomplete Gamma Functions of Complex Argument 109
3 Watson's Lemma 112
4 Airy Integral of Complex Argument; Compound Asymptotic Expansions 116
5 Ratio of Two Gamma Functions; Watson's Lemma for Loop Integrals 118
6 Laplace's Method for Contour Integrals 121
7 Saddle Points 125
8 Examples 127
9 Bessel Functions of Large Argument and Order 130
10 Error Bounds for Laplace's Method; the Method of Steepest Descents 135
Historical Notes and Additional References 137

5 Differential Equations with Regular Singularities; Hypergeometric and Legendre Functions

1	Existence Theorems for Linear Differential Equations: Real Variables	139
2	Equations Containing a Real or Complex Parameter	143
3	Existence Theorems for Linear Differential Equations: Complex Variables	145
4	Classification of Singularities; Nature of the Solutions in the Neighborhood of a Regular Singularity	148
5	Second Solution When the Exponents Differ by an Integer or Zero	150
6	Large Values of the Independent Variable	153
7	Numerically Satisfactory Solutions	154
8	The Hypergeometric Equation	156
9	The Hypergeometric Function	159
10	Other Solutions of the Hypergeometric Equation	163
11	Generalized Hypergeometric Functions	168
12	The Associated Legendre Equation ,	169
13	Legendre Functions of General Degree and Order	174
14	Legendre Functions of Integer Degree and Order	180
15	Ferrers Functions	185
	Historical Notes and Additional References	189

6 The Liouville-Green Approximation

1	The Liouville Transformation	190
2	Error Bounds: Real Variables	193
3	Asymptotic Properties with Respect to the Independent Variable	197
4	Convergence of $'f(F)$ at a Singularity	200
5	Asymptotic Properties with Respect to Parameters	v 203
6	Example:-Parabolic Cylinder Functions of Large Order	206
7	A Special Extension	208
8	Zeros	211
9	Eigenvalue Problems	214
10	Theorems on Singular Integral Equations	217
11	Error Bounds: Complex Variables	220
12	Asymptotic Properties for Complex Variables	223
13	Choice of Progressive Paths	224
	Historical Notes and Additional References	228

Differential Equations with Irregular Singularities; Bessel and Confluent Hypergeometric Functions

1	Formal Series Solutions	229
2	Asymptotic Nature of the Formal Series	232
3	Equations Containing a Parameter	236
4	Hankel Functions; Stokes' Phenomenon	237
5	The Function $7_s(z)$	241
6	Zeros of $J_v(z)$	244
7	Zeros of $7_v(z)$ and Other Cylinder Functions	248
8	Modified Bessel Functions	250
9	Confluent Hypergeometric Equation	254
10	Asymptotic Solutions of the Confluent Hypergeometric Equation	256
11	Whittaker Functions	260
12	Error Bounds for the Asymptotic Solutions in the General Case	262
13	Error Bounds for Hankel's Expansions	266
14	Inhomogeneous Equations	270
15	Struve's Equation	274
	Historical Notes and Additional References	277

8 **Sums and Sequences**

1	The Euler-Maclaurin Formula and Bernoulli's Polynomials	279
2	Applications	284
3	Contour Integral for the Remainder Term	289
4	Stirling's Series for In T(z)	293
5	Summation by Parts	295
6	Barnes'Integral for the Hypergeometric Function	299
7	Further Examples	302
8	Asymptotic Expansions of Entire Functions	307
9	Coefficients in a Power-Series Expansion; Method of Darboux	309
10	Examples	311
11	Inverse Laplace Transforms; Haar's Method	315
	Historical Notes and Additional References	321

9 Integrals: Further Methods

1	Logarithmic Singularities	322
2	Generalizations of Laplace's Method	325
3	Example from Combinatoric Theory	329
4	Generalizations of Laplace's Method (continued)	331

5 Examples 334
6 More General Kernels 336
7 Nicholson's Integral for$/_V^2(z) + K_v^2(z)$ 340
8 Oscillatory Kernels 342
9 Bleistein's Method 344
10 Example 346
11 The Method of Chester, Friedman, and Ursell 351
12 Anger Functions of Large Order 352
13 Extension of the Region of Validity 358
Historical Notes and Additional References 361

10 Differential Equations with a Parameter: Expansions in Elementary Functions

1 Classification and Preliminary Transformations 362
2 Case *V.* Formal Series Solutions 364
3 Error Bounds for the Formal Solutions 366
4 Behavior of the Coefficients at a Singularity 368
5 Behavior of the Coefficients at a Singularity (continued) 369
6 Asymptotic Properties with Respect to the Parameter 371
7 Modified Bessel Functions of Large Order 374
8 Extensions of the Regions of Validity for the Expansions of the Modified Bessel Functions 378
9 More General Forms of Differential Equation 382
10 Inhomogeneous Equations 386
11 Example: An Inhomogeneous Form of the Modified Bessel Equation 388
Historical Notes and Additional References 391

11 Differential Equations with a Parameter: Turning Points

1 Airy Functions of Real Argument 392
2 Auxiliary Functions for Real Variables 394
3 The First Approximation 397
4 Asymptotic Properties of the Approximation; Whittaker Functions with *m* Large 401
5 Real Zeros of the Airy Functions 403
6 Zeros of the First Approximation 405
7 Higher Approximations 408
8 Airy Functions of Complex Argument 413
9 Asymptotic Approximations for Complex Variables 416
10 Bessel Functions of Large Order 419

11 More General Form of Differential Equation 426
12 Inhomogeneous Equations 429
Historical Notes and Additional References 433

12 Differential Equations with a Parameter: Simple Poles and Other Transition Points

1 Bessel Functions and Modified Bessel Functions of Real Order and Argument 435
2 Case HI: Formal Series Solutions 438
3 Error Bounds: Positive f 440
4 Error Bounds: Negative £ 443
5 Asymptotic Properties of the Expansions 447
6 Determination of Phase Shift 449
7 Zeros 451
8 Auxiliary Functions for Complex Arguments 453
9 Error Bounds: Complex *u* and £ 457
10 Asymptotic Properties for Complex Variables 460
11 Behavior of the Coefficients at Infinity 462
12 Legendre Functions of Large Degree: Real Arguments 463
13 Legendre Functions of Large Degree: Complex Arguments 470
14 Other Types of Transition Points 474
Historical Notes and Additional References 478

13 Connection Formulas for Solutions of Differential Equations

1 Introduction 480
2 Connection Formulas at a Singularity 480
3 Differential Equations with a Parameter 482
4 Connection Formula for Case III 483
5 Application to Simple Poles 487
6 Example: The Associated Legendre Equation 490
7 The Gans-Jeffreys Formulas: Real-Variable Method 491
8 Two Turning Points 494
9 Bound States 497
10 Wave Penetration through a Barrier. I 501
11 Fundamental Connection Formula for a Simple Turning Point in the Complex Plane 503
12 Example: Airy's Equation 507
13 Choice of Progressive Paths 508
14 The Gans-Jeffreys Formulas: Complex-Variable Method 510
15 WavePenetrationthroughaBarrier.il 513
Historical Notes and Additional References 516

14 Estimation of Remainder Terms

1	Numerical Use of Asymptotic Approximations	519
2	Converging Factors	522
3	Exponential Integral	523
4	Exponential Integral (continued)	527
5	Confluent Hypergeometric Function	531
6	Euler's Transformation	536
7	Application to Asymptotic Expansions	540
	Historical Notes and Additional References	543

Answers **to Exercises** **545**

References **548**

Index of Symbols 561

General Index 563